

```
data2 <- read.csv("task-2.csv", header = TRUE)
```

```
names(data2)
```

```
head(data2)
```

```
str(data2)
```

```
X1 <- data2$X1
```

```
X2 <- data2$X2
```

```
X3 <- data2$X3
```

```
X4 <- data2$RESPONSE
```

```
> head(X1)
```

```
[1] "G_3" "G_3" "G_3" "G_3" "G_4" "G_3"
```

```
> head(X2)
```

```
[1] 15.2 12.4 15.2 19.0 12.3 12.1
```

```
> head(X3)
```

```
[1] 1.4 2.9 2.5 17.5 2.6 3.3
```

```
> head(X4)
```

```
[1] 9.5 11.7 9.6 6.6 13.8 11.5
```

```
library(moments)
```

```
summary_stats <- function(x) {
```

```
  c(
```

```
    mean = mean(x),
```

```
    median = median(x),
```

```
    sd = sd(x),
```

```
    skewness = skewness(x),
```

```
    kurtosis = kurtosis(x)
```

```
  )
```

```
}
```

```
stats <- data.frame(
```

```
  X2 = summary_stats(X2),
```

```
  X3 = summary_stats(X3),
```

```
  X4 = summary_stats(X4)
```

```
)
```

```
> print(stats)
```

	X2	X3	X4
mean	14.84100000	6.529667	9.4473333
median	14.95000000	4.450000	9.1000000
sd	2.76390114	5.963693	2.9037330
skewness	0.03638634	1.307374	0.4152476
kurtosis	1.95786725	4.008651	2.5936182

```
# Корелационни матрици (Pearson и Spearman)
cor_pearson <- cor(stats, method = "pearson")
cor_spearman <- cor(stats, method = "spearman")
```

```
> cor_pearson
      X2      X3      X4
X2 1.0000000 0.5657396 0.9938800
X3 0.5657396 1.0000000 0.6408993
X4 0.9938800 0.6408993 1.0000000
```

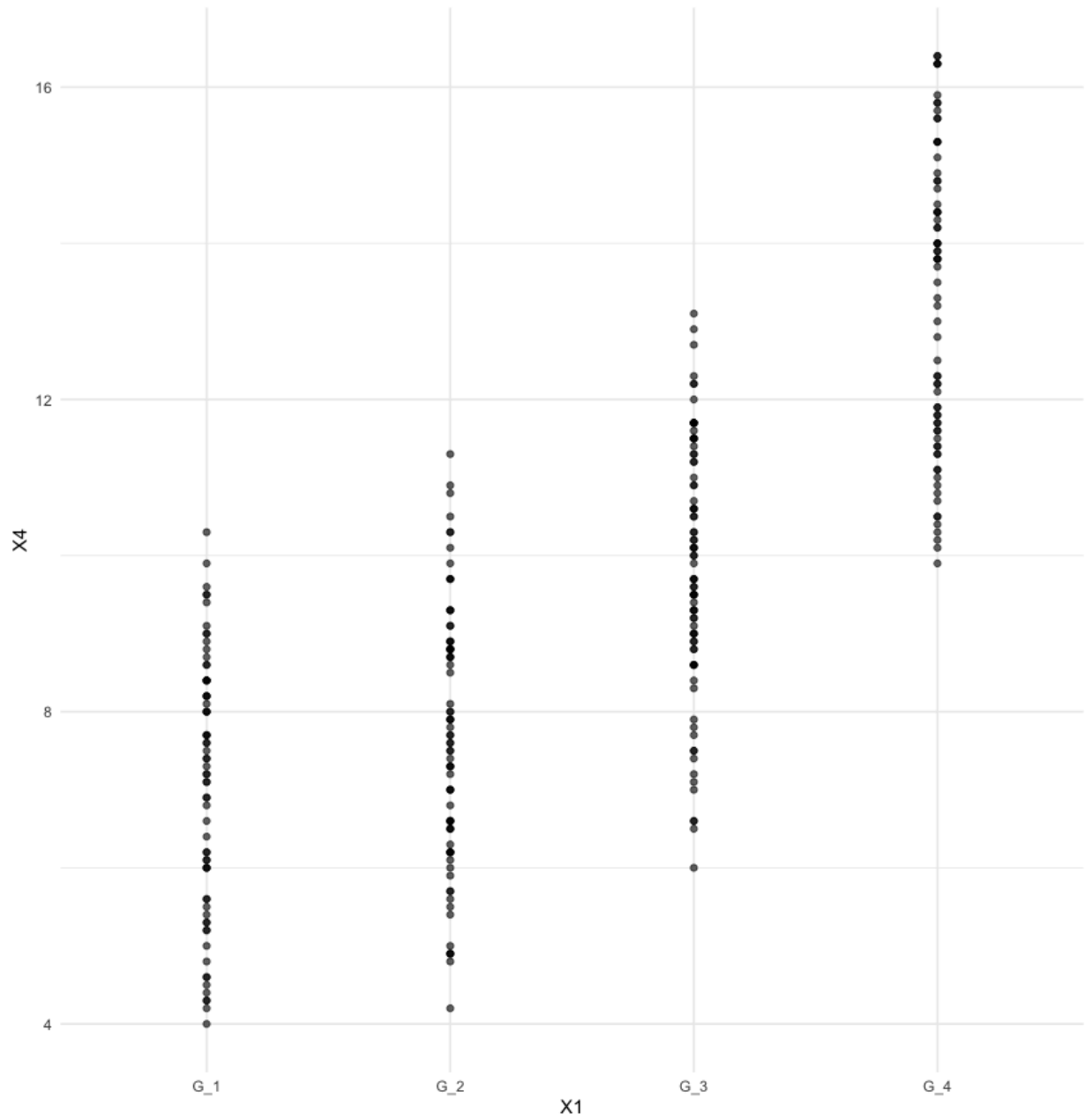
```
> cor_spearman
      X2 X3 X4
X2 1.0 0.7 0.9
X3 0.7 1.0 0.9
X4 0.9 0.9 1.0
```

От тук можем да видим кои двойки имат по-слаб или по-силен корелационен коефициент:

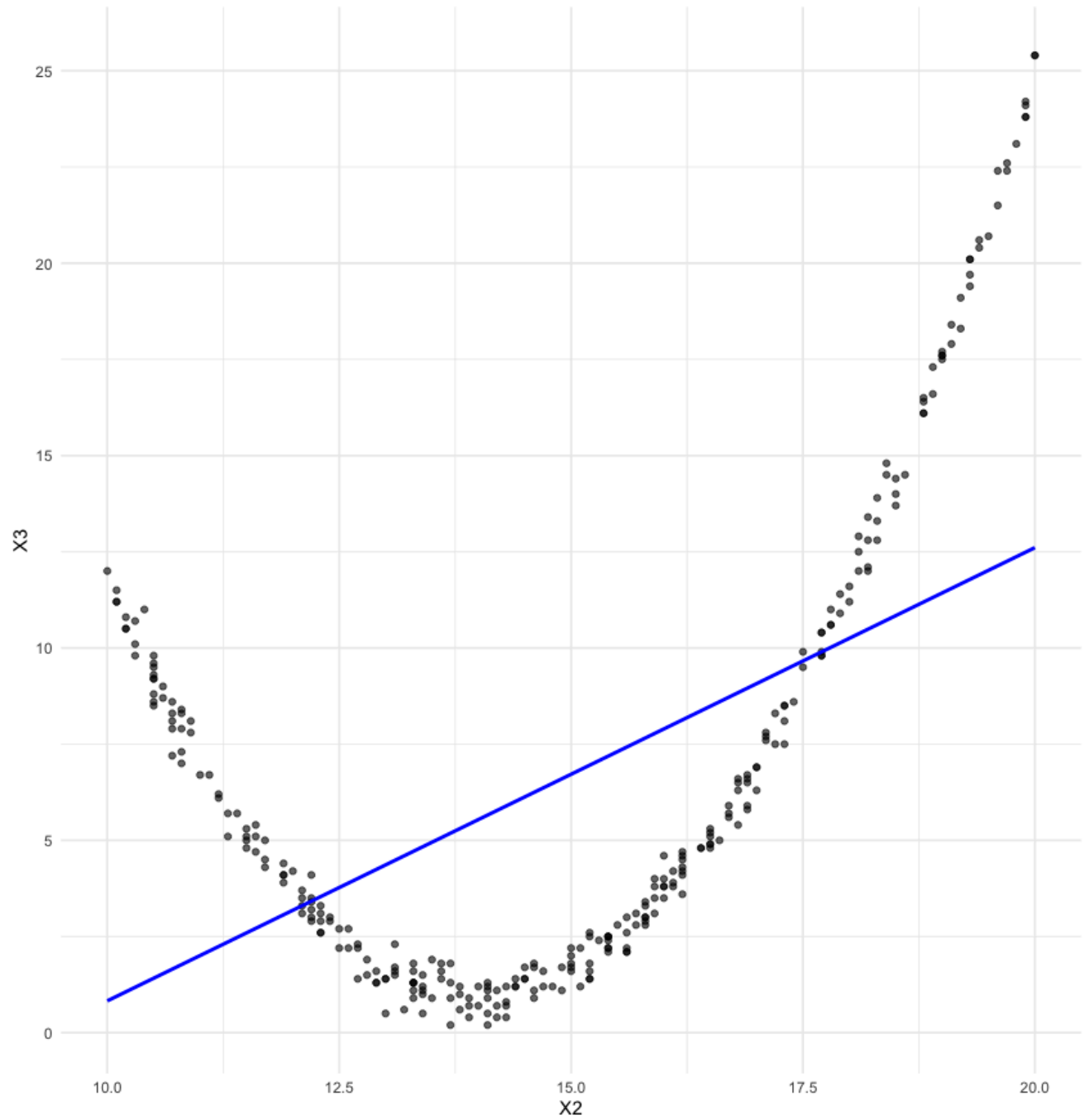
- **Корелационен коефициент > 0.7:** силна връзка.
- **0.4–0.7:** умерена.
- **< 0.4:** слаба.

```
# 3a. Scatterplots с плътност на точките (ggplot2)
library(ggplot2)
pairs_plot <- function(a, b) {
  ggplot(data2, aes_string(x = a, y = b)) +
    geom_point(alpha = 0.6) +
    geom_smooth(method = "lm", se = FALSE, color = "blue") +
    theme_minimal()
}
```

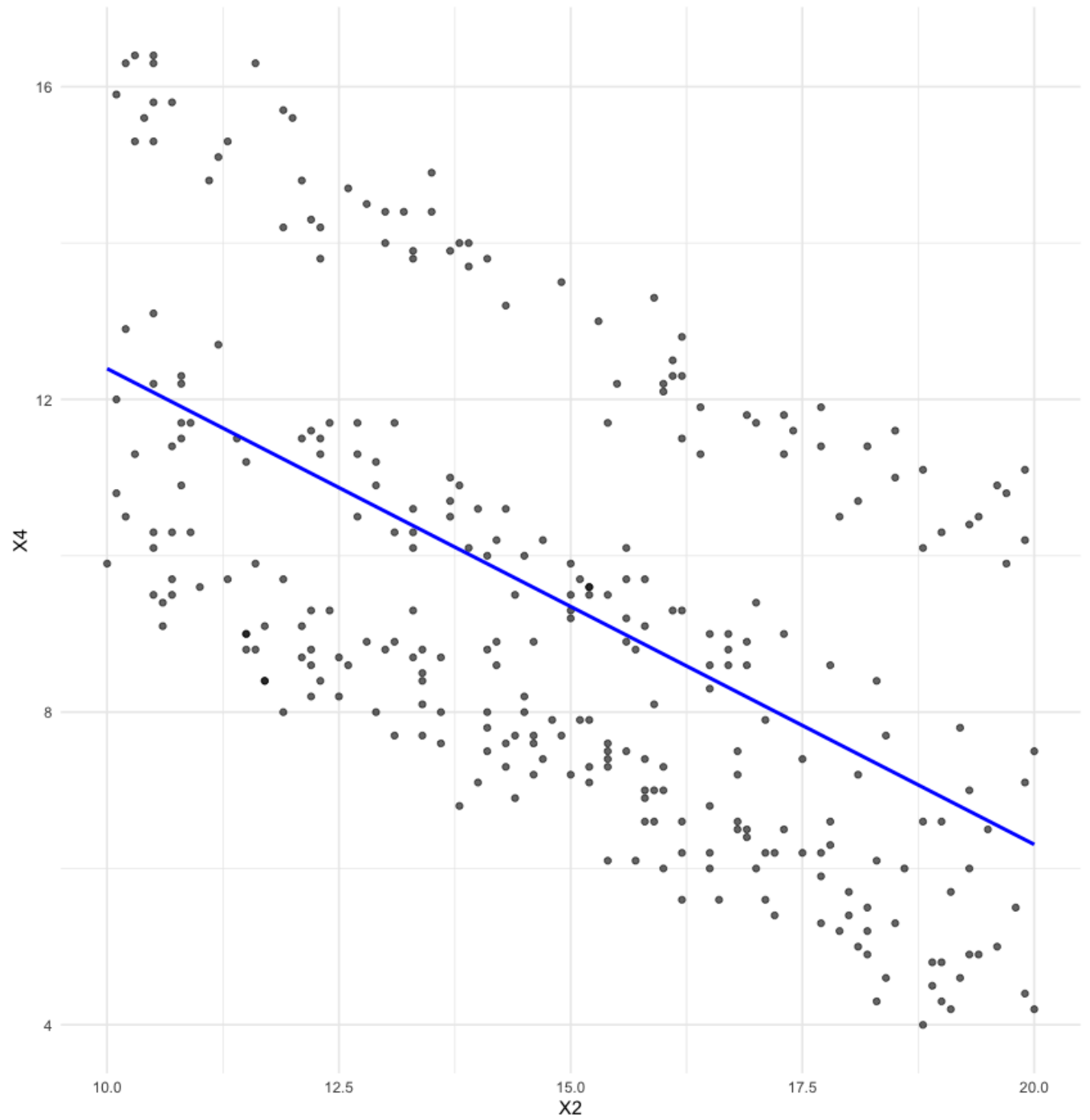
```
# X1 vs X4  
pairs_plot("X1", "X4")
```



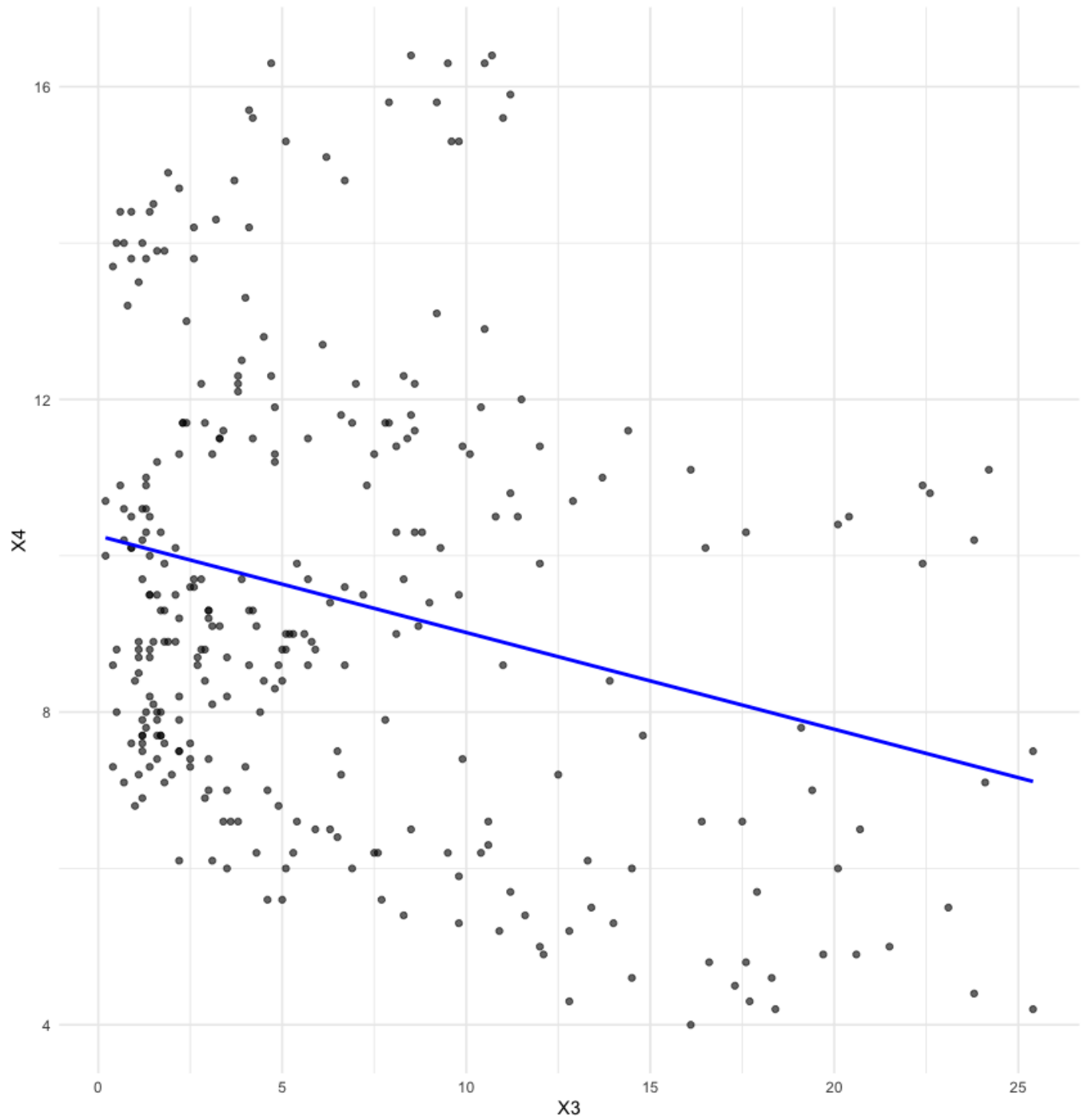
```
# X2 vs X3  
pairs_plot("X2", "X3")
```



```
# X2 vs X4  
pairs_plot("X2", "X4")
```



```
# X3 vs X4  
pairs_plot("X3", "X4")
```



```
# Проверка за нужда от трансформации  
library(MASS)
```

```
# Например за X2 vs X4:
```

```
bc <- boxcox(lm(X4 ~ X2, data = data2), plotit = FALSE)
```

```
lambda_opt <- bc$x[which.max(bc$y)]
```

```
> lambda_opt
```

```
[1] 0.2
```

```
X4_bc <- (X4^lambda_opt - 1) / lambda_opt
```

```
> head(X4_bc)
```

```
[1] 2.843587 3.177248 2.860031 2.292540 3.451734 3.149099
```

```
> cor(data2$X2, data2$RESPONSE)
```

```
[1] -0.5792863
```

```
> cor(data2$X2, X4_bc)
```

```
[1] -0.5976935
```

```
За X2 и X3
```

```
Най-силна корелация без трансформиране на данни
```

```
> cor(data2$X2, data2$X3)
```

```
[1] 0.5458884
```

```
> cor(log(X2), X3)
```

```
[1] 0.4668114
```

```
> cor(X2, sqrt(X3))
```

```
[1] 0.4866597
```

```
Корелацията намалява след двете трансформации
```